$$\mathcal{A} = \{000, 011\}, \quad H = \langle 011 \rangle$$

OBSERVATION	000	011	GAIN	FIBER
Read a	0	0	0 bits	2 remain
Read b	0	1	1 bit	1 remains
Read (b,b)	(0,0)	(1,1)	1 bit	1 remains

$\text{gain} = \text{rank}(Q|_H)$, not the number of output coordinates.

Bits refer to the uniform distribution on the two compatible realizations.